

MODEL NO : ET050FW04-T

MODEL VERSION: 01

SPEC VERSION : 1.0

ISSUED DATE: 2023-01-03

☐ Preliminary Specification

☒ Final Product Specification

Customer : _____

Approved by	Notes

Youri Confirmation

Prepared by	Reviewed by	Approved by
John	Johnny	Wang

1. Introduction

1.1 Scope of application

LCD specification: Dots 480xRGBx854

As to basic specification of the driver IC, refer to the IC (GalsxyCore:GC9503V) specification and data book.

All material & processing of the LCD module should be Lead Free.

1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL;

ALL Viewing Type LCD

480 dot-segment and 854 dot-common outputs;

16.7M Color can be selected by software;

White LED back light;

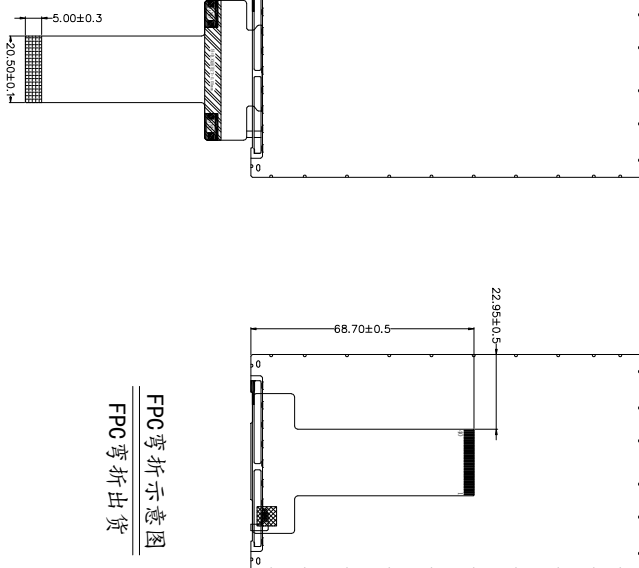
2lane MIPI interface

1.3 Applications:

2. LCM General specification

ITEM	Standard value	Unit
LCD Type	Normally Black	--
Drive element	TFT active matrix	--
Number of pixels	480*3RGB(H)X854(V)	Dots
Pixel arrangement	RGB Vertical Stripe	--
Pixel Pitch (W*H)	0.1284 (H) x 0.1284 (V)	mm
Active area	61.63(H) x 109.65(V)	mm
Viewing direction	ALL Viewing	--
TFT Driver IC	GC9503V	--
TFT interface	2lane MIPI Interface	--
LCM Size(W*H*T)	66.40(W) ×120.00(H) ×2.05(T)	mm
Approx. Weight	TBD	g
Touch structure	--	--
Touch Driver IC	--	--
Touch Interface	--	--

1	Pin	SVOL	SVOL
2		DO P	
3		DO N	
4	GND		
5	DI P		
6	DI N		
7	GND		
8	CLK P		
9	CLK N		
10	GND		
11	NC		
12	NC		
13	GND		
14	NC		
15	NC		
16	GND		
17	GND		
18	V10L 8V		
19	V10L 8V		
20	NC 7E		
21	NC		
22	NC 1D		
23	NC		
24	RESST		
25	NC		
26	NC		
27	GND		
28	LEAK		
29	LEAK		
30	GND		
31	NC		
32	GND		
33	GND		
34	NC		
35	LEDA		
36	LEDA		
37	GND		
38	VCC2 8V		
39	VCC2 8V		
40	NC		



FPC弯折示意图

FPC弯折出货

1	Pin	SVOL	SVOL
2		DO P	
3		DO N	
4	GND		
5	DI P		
6	DI N		
7	GND		
8	CLK P		
9	CLK N		
10	GND		
11	NC		
12	NC		
13	GND		
14	NC		
15	NC		
16	GND		
17	GND		
18	V10L 8V		
19	V10L 8V		
20	NC 7E		
21	NC		
22	NC 1D		
23	NC		
24	RESST		
25	NC		
26	NC		
27	GND		
28	LEAK		
29	LEAK		
30	GND		
31	NC		
32	GND		
33	GND		
34	NC		
35	LEDA		
36	LEDA		
37	GND		
38	VCC2 8V		
39	VCC2 8V		
40	NC		

A schematic diagram of a 6-LED string. It consists of a horizontal line with six LEDs connected in series. Each LED is represented by a triangle pointing to the right with two short lines extending from its base. A resistor, represented by a zigzag line, is connected in series with the LEDs. The input terminal on the left is labeled 'LED+' and the output terminal on the right is labeled 'LED-'. The resistor is labeled '100 ohm'.

LED CIRCUIT DIAGRAM

- NOTES:
1. DISPLAY TYPE: 5.0", 480x854 TFT LCD
 2. DISPLAY MODE: transmissive **Normally Black**
 3. VIEWING DIRECTION: **ALL**
 4. DRIVER IC: GC9503V
 5. LCM (White 9 AVG 1/9) :
Brightness: 500cd/m² (TPP)
Uniformity: 80% (MIN)
Back Light: 12 chip white LEDs If=40mA, Vf=18V~19, 2V
7. OPERATING TEMP: -20° C to 60° C, STORAGE TEMP: -30° C to 70° C
8. * Critical Parameter, () ref Parameter, [] cpk Parameter
Unspecified Tolerances: ±0.20mm
Modification mark:
9. SUGGESTION: TP window size unilateral increase 0.3~0.5mm than LCM A.A.
10. REQUIREMENTS ENVIRONMENTAL PROTECTION: RoHS

深圳市优瑞泰克科技有限公司									
PART NO.:		ET050FW04-T				MODULE NO.:		00	
DRAWN BY:		LJU		DATE:		PROJECTION:			
CHECKED BY:				DATE:		SCALE: N.T.S		UNIT: mm	
APPROVED BY:				DATE:		SHEET: 1 OF 1			

3.Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit
LCM Operating Temperature	T _{OPR}	-20	+60	°C
LCM Storage Temperature	T _{STG}	-30	+70	°C
Humidity	RH	--	90	%

4.Electrical Characteristics

4.1 TFT DC Characteristics

Characteristics	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for I/O	VDDIO	1.65	1.8	3.3	V
Supply Voltage for(DC/DC)	VDD	2.5	2.8	3.3	V
Supply Voltage for(DC/DC)	AVDD	--	--	--	V
Supply Voltage for(DC/DC)	AVEE	--	--	--	V

4.2 Back-Light Unit Characeristics

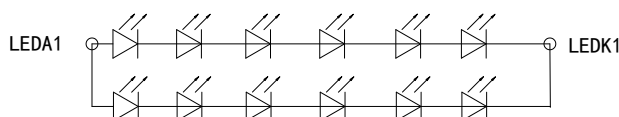
The back-light system is an edge-lighting type with 12 white LEDs. The characteristics of the back-light are shown in the following tables.

Characteristics	Symbol	Min.	Type	Max.	Unit	Notes
Forward Voltage	V _F	18	--	19.2	V	--
Forward current	I _F	--	40	--	mA	--
Luminance(With LCD)	L _v	--	500	--	cd/m ²	--
LED life time	N/A	--	30,000	--	Hr	Note 1

Note:

- (1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I_L=20mA/LED. The LED life time could be decreased if operating I_L is larger than 25mA/LED.

Backlight circuit diagram shown in below:



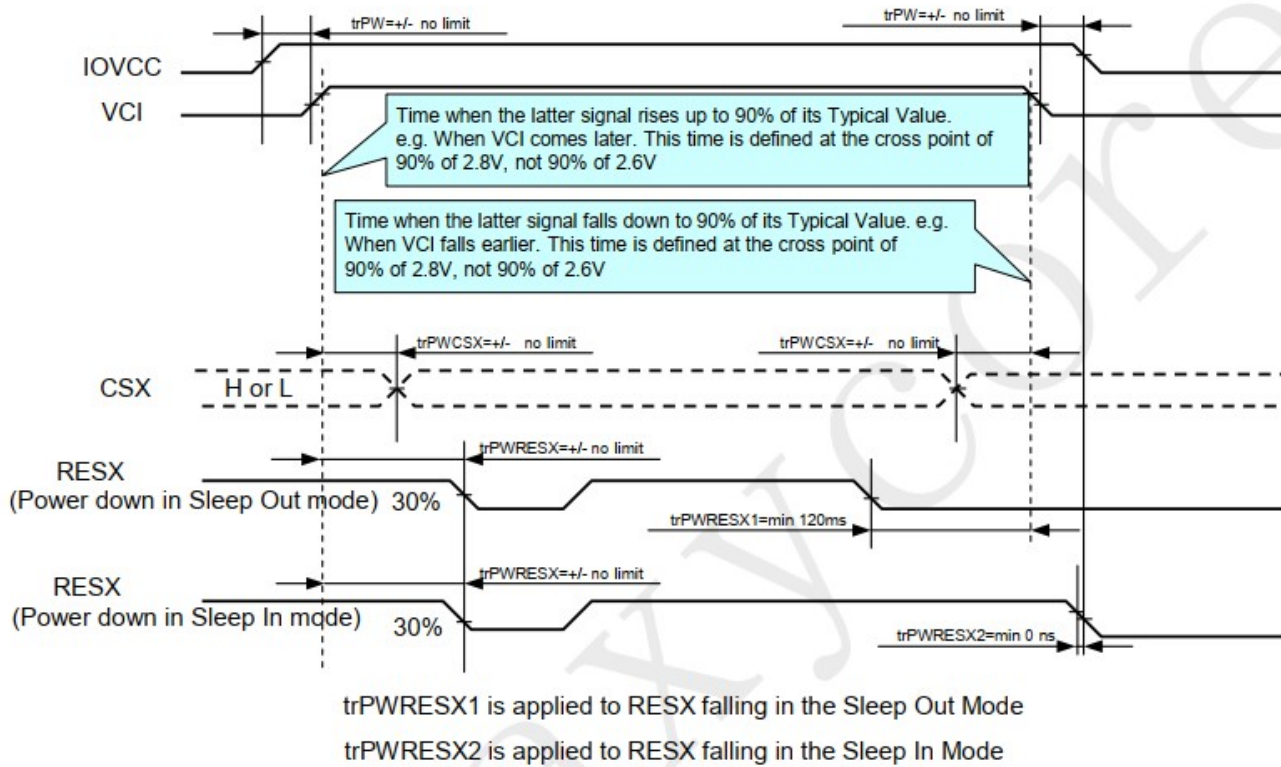
5. Module Function Description

Pin No.	Symbol	Functional	Notes
1	GND	Power Ground	
2	MIPI_0P	MIPI-DSI clock data Lane 0 pegative-end input/output pin.	
3	MIPI_0N	MIPI-DSI clock data Lane 0 nositive-end input/output pin.	
4	GND	Power Ground	
5	MIPI_1P	MIPI-DSI clock data Lane 0 pegative-end input/output pin.	
6	MIPI_1N	MIPI-DSI clock data Lane 0 nositive-end input/output pin.	
7	GND	Power Ground	
8	MIPI_CKP	MIPI-DSI clock data Lane pegative-end input pin.	
9	MIPI_CKN	MIPI-DSI clock data Lane nositive-end input pin.	
10	GND	Power Ground	
11~12	NC	NC	
13	GND	Power Ground	
14~15	NC	NC	
16~17	GND	Power Ground	
18~19	IOVCC_1.8V	I/O Voltage (VDDI to DGND): 1.65V ~ 3.3V	
20	NC/TE	NC	
21	NC	NC	
22	NC/ID	NC	
23	NC	NC	
24	RESET	Reset signal input terminal. Active at 'L'.	
25~26	NC	NC	
27	GND	Power Ground	
28~29	LEDK	Power supply for backlight cathode input terminal.	
30	GND	Power Ground	
31	NC	NC	
32~33	GND	Power Ground	
34	NC	NC	
35~36	LEDA	Power supply for backlight anode input terminal.	
37	GND	Power Ground	
38~39	VCC_2.8V	Power supply : 2.5V ~ 3.3V	
40	NC	NC	

6. Timing Characteristics

Case 1 –RESX line is held High or Unstable by Host at Power ON

If the RESX line is held high or unstable by the host during Power On, then a Hardware Reset must be applied after both VCI and IOVCC have been applied – otherwise correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.



DSI-MIPI Interface Timing Characteristics of IC

Normal Write Mode (VCC = IOVCC=2.4~3.3V)

High Speed Mode – Clock Channel Timing

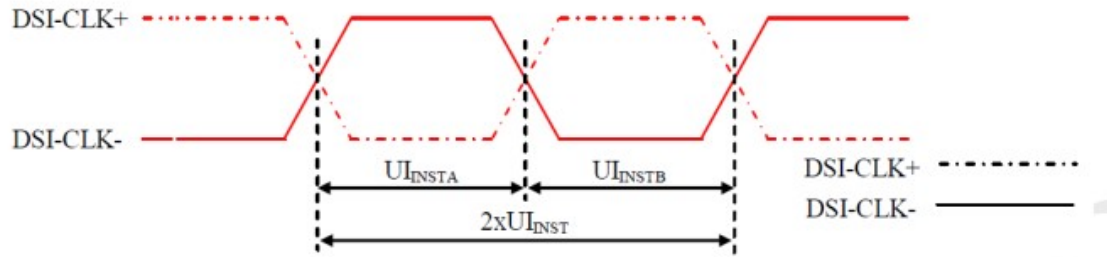


Figure 114 DSI Clock Channel Timing

Table 45 DSI Clock Channel Timing

Signal	Symbol	Parameter	Min	Max	Unit
DSI-CLK+/-	$2xUI_{INST}$	Double UI instantaneous	4	25	ns
DSI-CLK+/-	UI_{INSTA}, UI_{INSTB}	UI instantaneous Half	2	12.5	ns

Note: $UI = UI_{INSTA} = UI_{INSTB}$

High Speed Mode – Data Clock Channel Timing

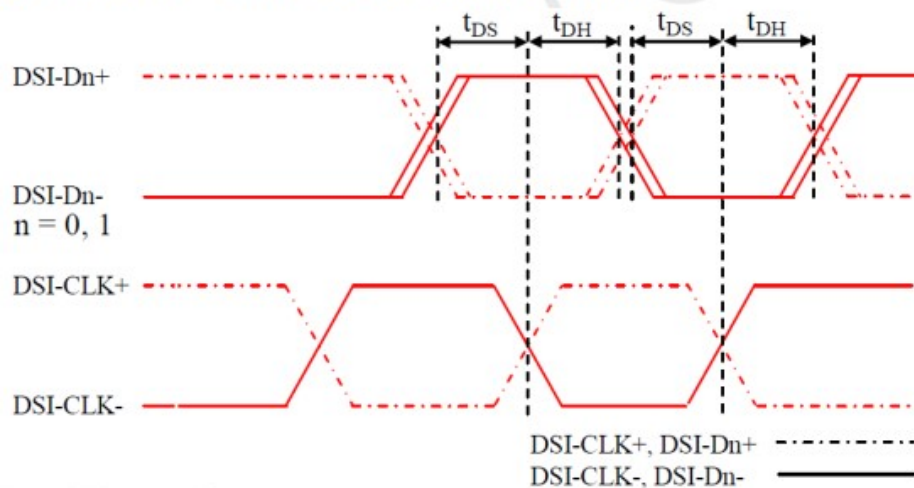


Figure 115 DSI Data to Clock Channel Timings

Table 46 DSI Data to Clock Channel Timings

Signal	Symbol	Parameter	Min	Max
DSI-Dn+/-, n=0 and 1	t_{DS}	Data to Clock Setup time	$0.15xUI$	-
	t_{DH}	Clock to Data Hold Time	$0.15xUI$	-

High Speed Mode – Rise and Fall Timings

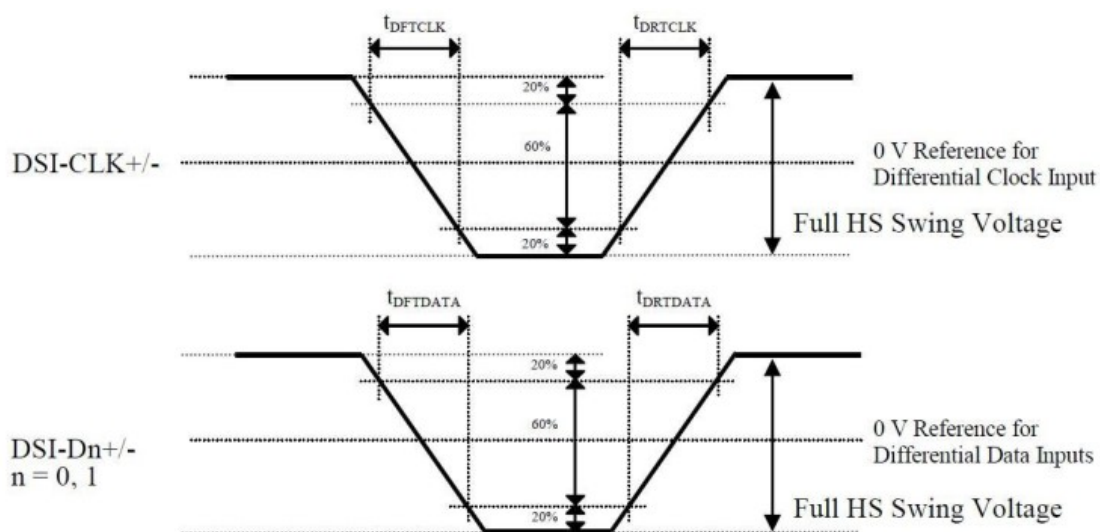


Figure 116 Rise and Fall Timings on Clock and Data Channels

Table 47 Rise and Fall Timings on Clock and Data Channels

Parameter	Symbol	Condition	Specification			Unit
			Min	Typ	Max	
Differential Rise Time for Clock	t_{DRTCLK}	DSI-CLK+/-	-	-	150 (Note)	ps
Differential Rise Time for Data	$t_{DRTDATA}$	DSI-Dn+/- n=0 and 1	-	-	150 (Note)	ps
Differential Fall Time for Clock	t_{DFTCLK}	DSI-CLK+/-	-	-	150 (Note)	ps
Differential Fall Time for Data	$t_{DFTDATA}$	DSI-Dn+/- n=0 and 1	-	-	150 (Note)	ps

Note: The display module has to meet timing requirements, what are defined for the transmitter (MPU) on MIPI D-Phy standard

Low Speed Mode – Bus Turn Around

Lower Power Mode and its State Periods are illustrated for reference purposes on the Bus Turnaround (BTA) from the MPU to the Display Module (GC9503V) sequence below.

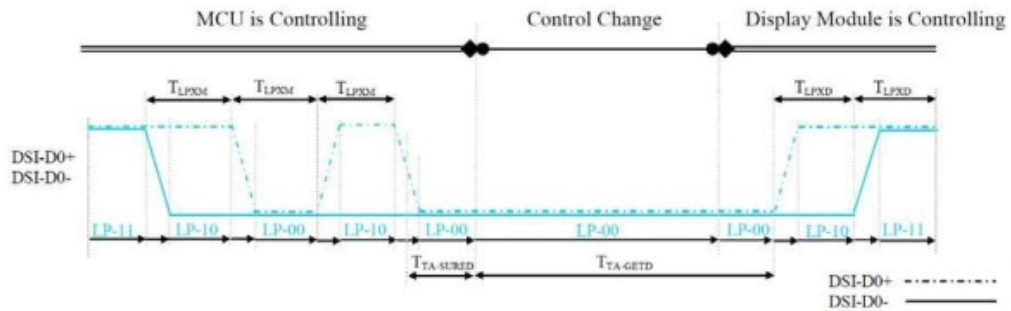


Figure 117 BTA from the MPU to the Display Module

Lower Power Mode and its State Periods are illustrated for reference purposes on the Bus Turnaround (BTA) from the Display Module (GC9503V) to the MPU sequence below.

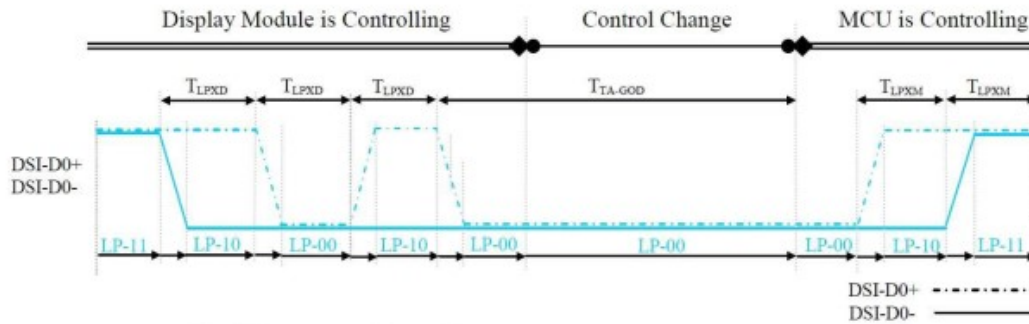


Figure 118 BTA from the Display Module to the MPU

Table 48 Low Power State Period Timings – A

Signal	Symbol	Description	Min	Max	Unit
DSI-D0+/-	T_{LPXM}	Length of LP-00, LP-01, LP-10 or LP-11 periods MPU $\rightarrow$ Display Module (GC9503V)	50	75	ns
DSI-D0+/-	T_{LPXD}	Length of LP-00, LP-01, LP-10 or LP-11 periods Display Module (GC9503V) $\rightarrow$ MPU	50	75	ns
DSI-D0+/-	$T_{TA-SURED}$	Time-out before the Display Module (GC9503V) starts driving	T_{LPXD}	$2 \times T_{LPXD}$	ns

Table 49 Low Power State Period Timings – B

Signal	Symbol	Description	Time	Unit
DSI-D0+/-	$T_{TA-GETD}$	Time to drive LP-00 by Display Module (GC9503V)	$5 \times T_{LPXD}$	ns
DSI-D0+/-	$T_{TA-GOOD}$	Time to drive LP-00 after turnaround request – MPU	$4 \times T_{LPXD}$	ns

Data Lanes from Low Power Mode to High Speed Mode

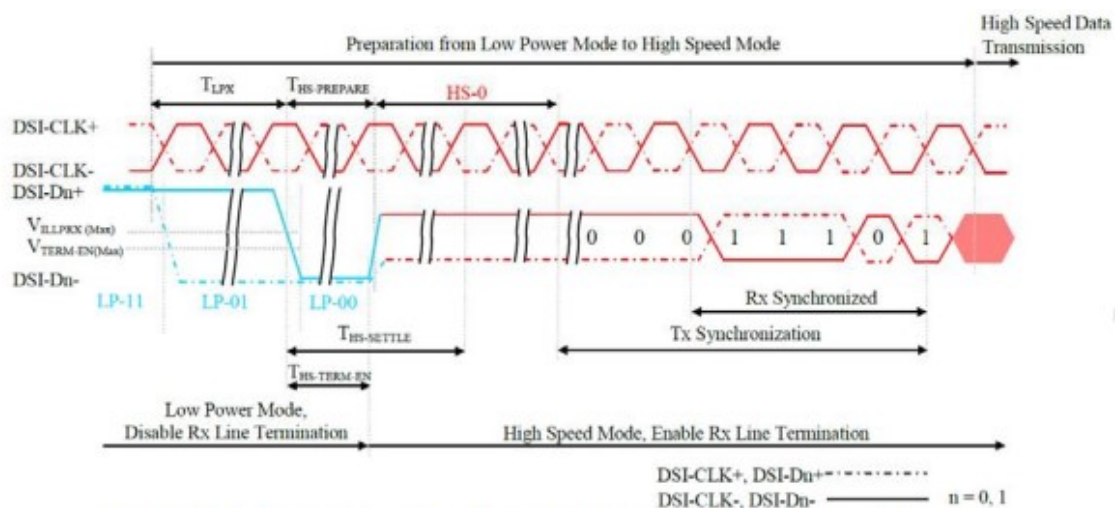


Figure 119 Data Lanes – Low Power Mode to High Speed Mode Timings

Table 50 Data Lanes – Low Power Mode to High Speed Mode Timings

Signal	Symbol	Description	Min	Max	Unit
DSI-Dn+/-, n=0 and 1	T_{LPX}	Length of any Low Power State Period	50	-	ns
DSI-Dn+/-, n=0 and 1	$T_{HS-PREPARE}$	Time to drive LP-00 to prepare for HS Transmission	$40+4xUI$	$85+6xUI$	ns
DSI-Dn+/-, n=0 and 1	$T_{HS-TERM-EN}$	Time to enable Data Lane Receiver line termination measured from when Dn crosses VILMAX	-	$35+4xUI$	ns

Data Lanes from High Speed Mode to Low Power Mode

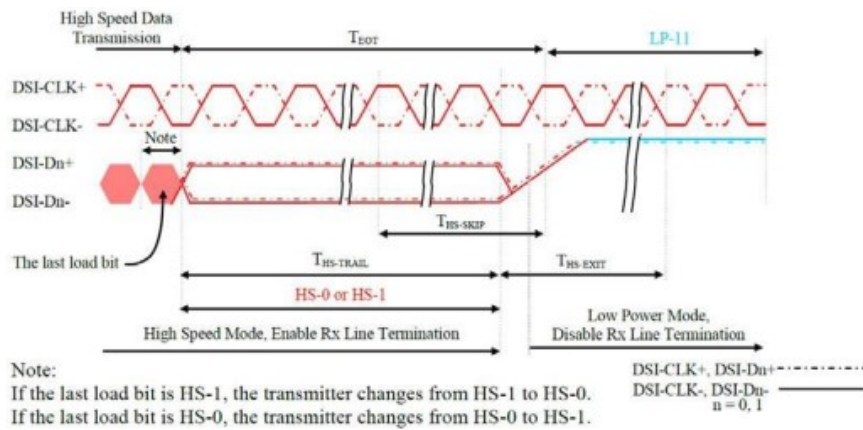


Figure 120 Data Lanes – High Speed Mode to Low Power Mode Timings

Table 51 Data Lanes – High Speed Mode to Low Power Mode Timings

Signal	Symbol	Description	Min	Max	Unit
DSI-Dn+/-, n=0 and 1	$T_{HS-SKIP}$	Time-Out at Display Module (GC9503V) to ignore transition period of EoT	40	$55+4xUI$	ns
DSI-Dn+/-, n=0 and 1	$T_{HS-EXIT}$	Time to driver LP-11 after HS burst	100	-	ns

DSI Clock Burst – High Speed Mode to/from Low Power Mode

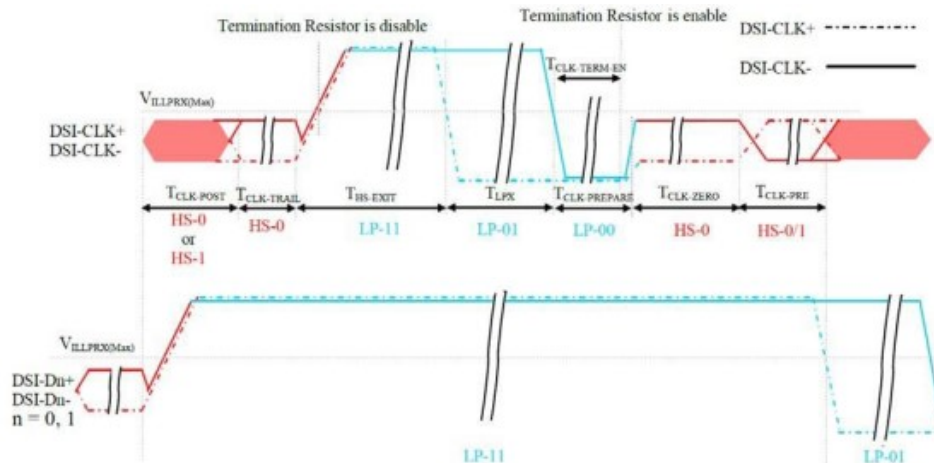


Figure 121 Clock Lanes – High Speed Mode to/from Low Power Mode Timings

Table 52 Clock Lanes – High Speed Mode to/from Low Power Mode Timings

Signal	Symbol	Description	Min	Max	Unit
DSI-CLK+/-	$T_{CLK-POST}$	Time that the MPU shall continue sending HS clock after the last associated Data Lanes has transitioned to LP mode	$60+52xUI$	-	ns
DSI-CLK+/-	$T_{CLK-TRAIL}$	Time to drive HS differential state after last payload clock bit of a HS transmission burst	60	-	ns
DSI-CLK+/-	$T_{HS-EXIT}$	Time to drive LP-11 after HS burst	100	-	ns
DSI-CLK+/-	$T_{CLK-PREPARE}$	Time to drive LP-00 to prepare for HS transmission	38	95	ns
DSI-CLK+/-	$T_{CLK-TERMEN}$	Time-out at Clock Lane to enable HS termination	-	38	ns
DSI-CLK+/-	$T_{CLK-PREPARE}$	Minimum lead HS-0 drive period before starting Clock	300	-	ns
DSI-CLK+/-	$T_{CLK-PRE}$	Time that the HS clock shall be driven prior to any associated Data Lane beginning the transition from LP to HS mode	$8xUI$	-	ns

Reset Description:

The MPU can inform to the display module that it should be reset in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode.

The Remote Application Reset (RAR) is using a following sequence:

Start: LP-11

Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)

• Mark-1: LP-00 =>LP-10 =>LP-11

End: LP-11

This sequence is illustrated for reference purposes below:

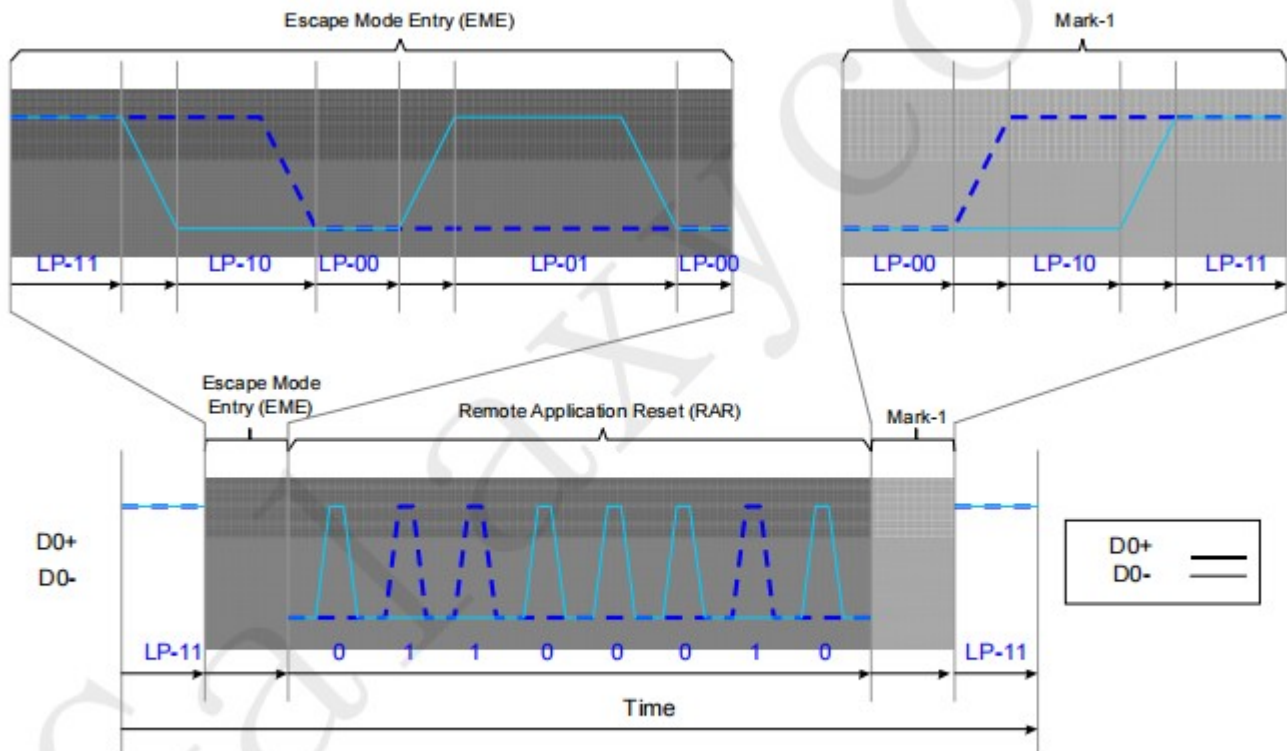


Figure 27 Remote Application Reset (RAR)

7. Optical Characteristics

Optical Specification

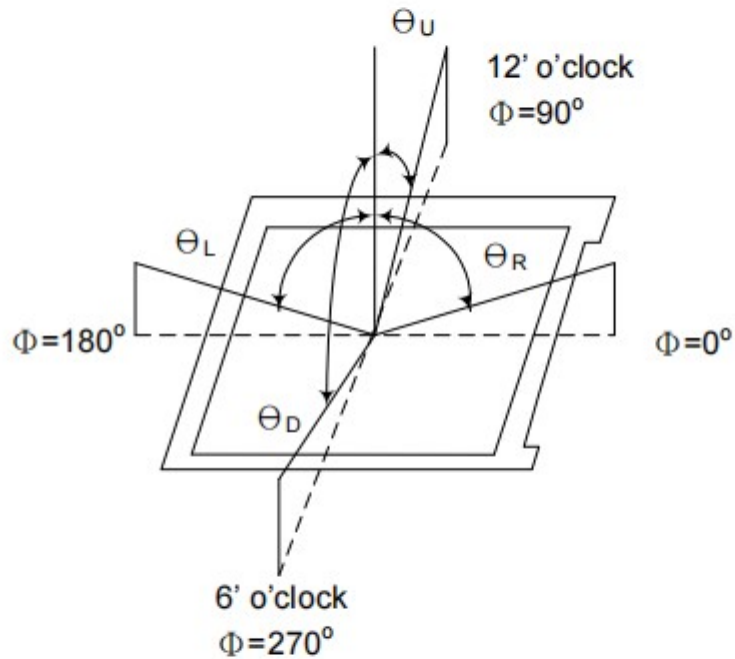
4.1 Optical specification

908OI908		Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Transmittance (with Polarizer)		T (%)	Θ=0 Normal viewing angle	—	4.47	—	%	Measuring with Polarizer , Reference Only
Transmittance (without Polarizer)		T (%)		—	14.2	—	%	
Contrast		CR		640	800	—	—	(1)(2)
Response time	Rising	T _R		—	16	21	msec	(1)(3)
	Falling	T _F		—	19	24		
Color gamut	(%)			—	70	—	%	C-light
Color chromaticity (CIE1931)	White	W _x	-0.02	0.305	+0.02	—	(1)(4) CF glass	
		W _y		0.340		—		
	Red	R _x		0.665		—		
		R _y		0.327		—		
	Green	G _x		0.276		—		
		G _y		0.594		—		
	Blue	B _x		0.134		—		
		B _y		0.118		—		
Viewing angle	Hor.	Θ _L	CR>10	—	80	—	—	(1)(4) Measuring with Polarizer , Reference Only
		Θ _R		—	80	—		
	Ver.	Θ _U		—	80	—		
		Θ _D		—	80	—		
Optima View Direction		Free						(5)

Measuring Equipment

- FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics.

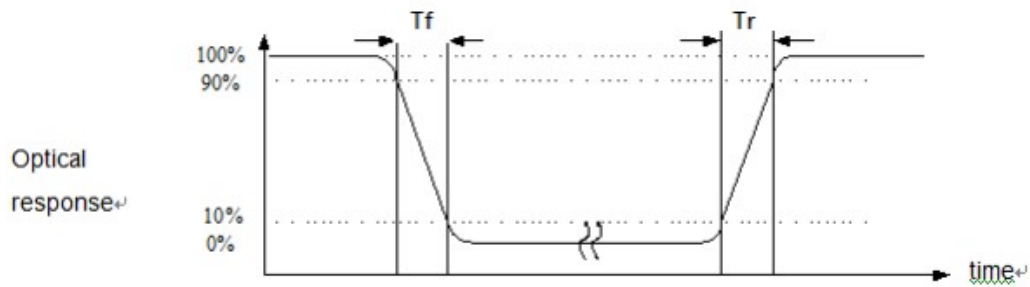
Note (1) Definition of Viewing Angle:



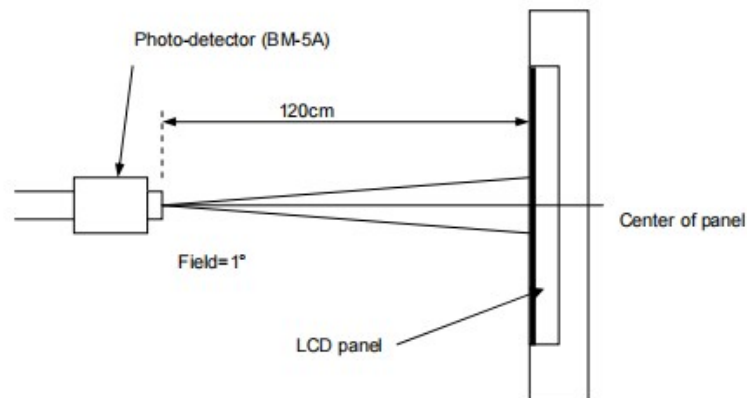
Note (2) Definition of Contrast Ratio (CR) :
measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

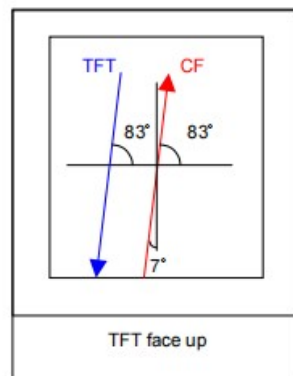
Note (3) Definition of Response Time : Sum of T_R and T_F



Note (4) Definition of optical measurement setup



Note (5) Rubbing Direction (The different Rubbing Direction will cause the different optima view direction).



8. Reliability Test Item

No.	Test Item	Test Condition	Notes
1	High Temp. Storage	+70°C / 96H	1. Functional test isOK. Missing Segment,short, unclear segment non-display,display abnormally and liquid crystal leakare un-allowed. 2. No low temperature bubbles,end seal loose andfall, frame rainbow.
2	Low Temp. Storage	-30°C / 96H	
3	High Tempe. Operating	+60°C / 96H	
4	Low Tempe. Operating	-20°C / 96H	
5	High Temperature /Humidity storage	50°C x 90%RH /96H	
6	Thermal and cold shock	Static state, -20°C (30min) ~60°C (30min) , 50 cycles	

9. Packing Method----TBD

- END -